

NATIONAL RESEARCH UNIVERSITY
HIGHER SCHOOL OF ECONOMICS

Faculty of Computer Science
Bachelor's Programme "Data Science and Business Analytics"

Research Project Report on the Topic:
Becoming an R&D Engineer.
Understanding and Improving the Robustness of YOLO Object Detection Models
Under Low-Light Conditions

Fulfilled by:

Student of the Group БИАД246
Tikhonravov Vladislav Dmitrievich

01.06.2026

(date)

Assessed by the Project Supervisor:

Grinberg Petr Markovich
Visiting Lecturer
Faculty of Computer Science, HSE University

Contents

Annotation	4
1 Introduction	5
2 Related Works	7
3 Analysis of failure modes	9
3.1 Baseline Model	9
3.2 Experimental setup	10
3.2.1 Evaluation Metrics	10
3.2.2 Controlled Low-Light Degradation	11
3.3 Results	12
4 Methodology	14
4.1 Curriculum-Based Robustness Training (CBRT)	14
4.2 Illumination-Conditioned Confidence Calibration (ICCC)	15
4.3 Comparative and Combined Evaluation	16
5 Configuration for CBRT and ICCC	18
5.1 CBRT Training Configuration	18
5.2 ICCC Configuration	19
5.2.1 Automatic γ estimation	19
5.2.2 Calibration of coefficient k	20
5.3 Combined CBRT and ICCC setup	21
6 Results	23
6.1 CBRT and naive augmentation performance	23
6.2 ICCC performance	25
6.3 Combined CBRT and ICCC	27
6.4 Comparison with Existing Methods	27
6.4.1 Direct Comparison with Enhancement-Based Preprocessing	28
6.4.2 Direct Comparison with a Training-Time Method (MAET)	30
6.4.3 Contextual Comparison with Training-Time Methods	32
7 Conclusion	34

8	Use of generative AI	36
	References	38

Abstract

Modern object detection models based on deep learning demonstrate efficient performance results under standard imaging conditions, however, they still perform poorly in challenging illumination environments. Low-light conditions, in particular, typically lead to substantial degradation in terms of detection quality, resulting in missed objects and localization errors. In this work, we analyze the robustness of YOLO-based object detection models under reduced illumination to identify typical failure modes that arise when lighting conditions deteriorate. Based on the performed analysis, we further explore and implement approaches that are aimed to mitigate the identified failure modes and improve detection performance of the model in such conditions.

Аннотация

Современные модели обнаружения объектов, основанные на глубоком обучении демонстрируют эффективные результаты в стандартных условиях получения изображения, однако они по-прежнему плохо работают в сложных условиях освещения. В частности, условия слабого освещения обычно приводят к существенному снижению качества обнаружения, выражающемуся в пропущенных объектах и ошибках локализации. В данной работе проводится анализ устойчивости моделей обнаружения объектов на основе YOLO в условиях слабого освещения с целью выявления типичных режимов отказа, появляющихся при ухудшении условий освещения. На основе проведённого анализа далее рассматриваются и реализуются подходы направленные на смягчение выявленных режимов отказа и улучшение качества обнаружения модели в условиях слабой освещённости.

Keywords

Deep Learning, Object Detection, Computer Vision, Failure Modes, Robustness, YOLO

1 Introduction

In recent years, object detection has become one of the core fields in computer vision because of its wide range of real-world applications, such as video surveillance [22], autonomous driving [5], intelligent monitoring systems [1], and other fields. Modern object detection models based on deep learning demonstrate strong performance on standard benchmark datasets [14, 9]. In particular, YOLO-based [19] models [4, 13] achieve state-of-the-art detection accuracy on widely used benchmarks such as MS COCO [14] and PASCAL VOC [9].

Despite their success on standard well-lit benchmark datasets [14, 9], object detection models remain vulnerable to distribution shifts and adverse environmental variations [17]. Among such variations, illumination degradation represents one of the most common factors which affect detection performance. In real-world scenarios images can be frequently captured under low-light conditions, such as nighttime environments, indoor locations with insufficient lighting, or various weather conditions. Under such degraded lighting scenarios, the performance of object detection models diminishes substantially. Several empirical studies [7, 25, 15] report a substantial drop in mAP for YOLO-based models when evaluated on the low-light datasets [15, 18].

Considerable research effort has been devoted to mitigating the negative impact of low-light conditions on object detection performance. Existing approaches include low-light image enhancement as a preprocessing step [25] as well as illumination-aware data augmentation strategies that explicitly simulate lighting variations during training [21]. However, despite these efforts, the observed performance gap between standard and low-light conditions remains significant, which indicates that the current methods are still insufficient to fully address the problem and a deeper understanding of model behavior under low-light conditions is required.

The main goal of this work is to analyze the robustness of YOLO-based object detection models under illumination degradation. The study focuses on identifying and categorizing typical failure modes that occur when illumination conditions deteriorate. Through a detailed analysis of model performance on both standard and artificially degraded low-light images, this work aims to understand the underlying causes of performance degradation.

Unlike prior studies that evaluate existing enhancement methods [25] or apply generic illumination-based augmentation strategies [21], our work follows an analysis-based approach: at first we analyze the behavior of the detector under controlled illumination degradation, and then design targeted mitigation strategies motivated directly by the identified failure modes. Specifically, we introduce a decision-level calibration method that compensates for illumination-caused confidence suppression, and a curriculum-based training strategy that addresses representation-

level degradation. The results of this study are expected to improve the robustness and overall detection performance of the YOLO-based model in low-light environments.

2 Related Works

The YOLO (You Only Look Once) [19] family of models introduced a unified one stage object detection framework, where the separate components of object detection are unified into a single neural network. This approach substantially simplified the detection pipeline and allowed real-time object detection with competitive accuracy in comparison with traditional two-stage detectors [20]. The original YOLO model demonstrated that treating object detection as a regression problem is an efficient approach. The subsequent developments [4, 13] of YOLO architecture improved detection accuracy, robustness and computational efficiency. For example, YOLOv4 [4] incorporated a carefully designed backbone network CSPDarknet53 [4, 24], which improved gradient flow, feature reuse and reduced computational redundancy. Additionally, advanced data augmentation strategies such as Mosaic augmentation [12] and Self-Adversarial Training [4] were introduced, which increased the robustness of the model. These modifications allowed YOLOv4 to achieve state-of-the-art performance on standard benchmarks like MS COCO, while maintaining real-time inference speed [4]. As a result, YOLO became one of the most widely adopted object detection frameworks in real-life applications because of both accuracy and computational efficiency.

Studies show that YOLO-based detectors demonstrate strong performance on standard object detection benchmarks, however their accuracy significantly degrades when evaluated on the low-light datasets [4, 7]. The consistently reported performance degradation under low-light conditions compared to results obtained on well-lit benchmarks indicates a substantial robustness gap which occurs under illumination degradation conditions.

In order to mitigate the degradation of models in low-light environments, various approaches were developed. A common strategy is to apply low-light image enhancement as a preprocessing step before using the detector [23]. This approach aims to improve visibility for the model by increasing brightness and contrast and suppressing noise, which leads to improving downstream detection quality in practice. Several representative methods have been proposed within this approach. Zero-DCE [11] formulates light enhancement as a task of image-specific curve estimation using a lightweight network trained without any paired or unpaired reference data. SCI [16] develops a self-calibrated illumination learning framework that achieves competitive enhancement quality with extremely low computational cost through a weight-sharing cascaded architecture that requires only a single basic block for inference. In contrast to these learning-based methods, CLAHE [26] is a classical contrast-enhancement technique that applies histogram equalization locally over image tiles with a clipping limit to avoid noise amplification.

However, enhancement-based approach introduces artifacts and considerably increases the complexity of the system. Furthermore, empirical investigations [25] have shown that methods which enhance low-light images for visual quality might not necessarily improve performance, but can even harm it in high-level tasks such as object detection. This indicates that the relationship between illumination conditions and behavior of the detector is still insufficiently understood and analyzed in the literature.

Another approach targets low-light detection in an end-to-end manner by adopting illumination related modeling into the learning process. For example, MAET [7] introduces a self-supervised framework that explicitly models illumination-degrading transformations and uses them to improve object detection in dark environments. Recent works explore domain adaptation settings to transfer detectors trained on well-lit images to low-light conditions. Particularly, zero-shot day-night domain adaptation [8] focuses on generalizing a detector from well-lit domain scenarios to low-light ones without requiring real low-light training data. This demonstrates that illumination-invariant representation can substantially increase nighttime detection robustness.

Summarizing, despite considerable effort that has been devoted to improving object detection performance and robustness, most existing methods target the problem at the level of generic solutions, without first analyzing which specific failure modes dominate the detector’s behavior in low-light conditions. As a result, it remains unclear whether the proposed fixes address the actual underlying causes of degradation or simply reduce its symptoms. This motivates our work, which focuses on understanding and improving the robustness of YOLO-based object detection models under reduced illumination through systematic failure mode analysis and targeted model improvement strategies explicitly derived from that analysis.

3 Analysis of failure modes

3.1 Baseline Model

As a baseline detector, we select YOLOv8s from the Ultralytics implementation [13]. YOLOv8 model is a modern one-stage object detection framework that combines competitive detection accuracy with computational efficiency. We chose the “s” (small) configuration to achieve a balance between training efficiency and model capacity. Compared to the nano version (YOLOv8n) YOLOv8s offers a significantly larger feature extraction capacity, which is important for meaningful robustness analysis, but at the same time it remains lightweight computationally compared to larger configurations such as YOLOv8m, which makes it suitable for controlled experimentation within reasonable computational time.

The model was initialized with pretrained weights provided by the official Ultralytics implementation, which were originally trained on the MS COCO dataset [14]. Then we fine-tuned the model on the COCO 2017 train split under a controlled training configuration. While the pretrained weights already provide strong detection performance on COCO, performing an additional fine-tuning step under our own training configuration is necessary in order to ensure a fair comparison with the curriculum-based training strategy proposed in Section 4. Both the baseline and the curriculum-trained model must originate from the same training pipeline, so that any observed differences can be attributed to the curriculum strategy rather than to differences in training hyperparameters or augmentation policies.

Fine-tuning was performed on the full COCO 2017 train split (approximately 118,000 images) for 25 epochs, with a batch size of 16, an input image resolution of 640×640 , and the AdamW optimizer. A fixed random seed of 42 was used for reproducibility. All other hyperparameters follow Ultralytics default configuration [13].

The training procedure followed the default Ultralytics YOLOv8 pipeline without additional architectural modifications. The default Ultralytics augmentation pipeline was applied during training, including HSV (hue, saturation, value) jitter. We acknowledge that the default HSV jitter provides limited exposure to brightness variation; however, this augmentation is substantially weaker than our controlled low-light degradation approach; see Eq. (4).

The model we obtained serves as the reference detector for the following robustness evaluation under controlled low-light degradation.

3.2 Experimental setup

The experiments in this work are conducted on MS COCO dataset [14], one of the most commonly used benchmarks in the field of object detection. The dataset contains images of daily scenes with objects which belong to 80 pre-defined classes with high-quality bounding box annotations. We selected MS COCO because of its diversity of object categories, scene complexity, and its relevance as a benchmark in object detection.

3.2.1 Evaluation Metrics

To evaluate detection performance, we use a set of standard object detection metrics.

Precision measures the proportion of predicted bounding boxes that correspond to correct detections. It is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

where TP denotes true positives and FP denotes false positives.

Recall measures the proportion of ground-truth objects that are successfully detected by the model. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

where FN denotes false negatives.

mAP50, the Mean Average Precision at Intersection-over-Union(IoU) [20] threshold 0.5, evaluates detection quality by computing the average precision across all object classes using an IoU threshold of 0.5 to determine correct detections.

mAP50–95, the Mean Average Precision averaged over IoU thresholds from 0.5 to 0.95 with a step of 0.05, provides a stricter and more comprehensive measure of detection quality than mAP50. This metric reflects both classification accuracy and bounding box localization precision.

F1-score is a single-number summary that balances precision and recall. It is defined as:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

These metrics provide a comprehensive assessment of detection performance. Particularly, recall is sensitive to illumination degradation due to potential missed detections, while mAP50–95 reflects the impact of low-light conditions on localization accuracy.

3.2.2 Controlled Low-Light Degradation

In order to fairly evaluate the robustness of the baseline detector under illumination degradation, we introduce a controlled synthetic low-light transformation which we apply to the validation images. The goal of this approach is to simulate low-light conditions, while keeping all other factors unchanged. This allows us to isolate the effect of illumination degradation on detection performance.

Inspired by standard photometric intensity transformations and additive noise models widely adopted in image processing literature [10], we perform the degradation as a combination of gamma-based intensity transformation, global brightness attenuation, and additive Gaussian noise:

$$I_{\text{dark}} = \alpha \cdot I^\gamma + \epsilon \quad (4)$$

where I denotes the normalized input image in the range $[0, 1]$, γ controls non-linear intensity compression (gamma correction), α is a global brightness scaling factor, $\epsilon \sim \mathcal{N}(0, \sigma^2)$ represents additive Gaussian noise. Gamma correction with $\gamma > 1$ compresses higher intensity values and simulates reduced visibility in darker regions. The scaling factor $\alpha < 1$ further weakens the overall brightness. Additive Gaussian noise models sensor noise typically observed in low-light imaging scenarios.

To analyze robustness under varying illumination severity, we introduced three degradation levels in addition to the original (normal) validation set:

Table 3.1: Degradation parameters for each illumination condition.

Condition	Degradation Parameters		
	γ	α	σ
Normal	1	1	0
Mild	1.4	0.75	5
Medium	1.8	0.55	8
Severe	2.2	0.40	10

The parameters are selected to simulate a progressive increase in degradation severity. As the conditions become worse, γ increases, applying stronger non-linear compression that suppresses mid-tone and shadow detail. Simultaneously, α decreases, which reduces the overall brightness of the scene. Finally, σ increases, introducing more sensor noise, which is characteristic of low-light capture where camera sensors operate at higher gain settings. The Normal condition corresponds to an unmodified image ($\gamma = 1$, $\alpha = 1$, $\sigma = 0$).

The transformation was applied only to the validation images, while the bounding box annotations remained unchanged. This ensures that performance differences across illumination levels can be directly tied to changes in illumination characteristics rather than to dataset composition or annotation changes.

We refer to the original validation set as the Normal condition, and it serves as a reference baseline for comparison.

3.3 Results

To evaluate the robustness of the baseline detector under illumination changes, we evaluated the fine-tuned YOLOv8s model on four validation conditions: Normal, Mild, Medium, and Severe. The Normal condition corresponds to the original COCO validation split, while the remaining conditions were generated using the controlled low-light degradation approach from Eq. (4). Table 3.2 summarizes the detection metrics obtained.

Table 3.2: Baseline detection performance under different illumination conditions on COCO val2017.

Condition	Validation Metrics				
	Precision $\uparrow$	Recall $\uparrow$	mAP50 $\uparrow$	mAP50–95 $\uparrow$	F1 $\uparrow$
Normal	0.673	0.538	0.590	0.430	0.598
Mild	0.661	0.515	0.566	0.409	0.579
Medium	0.631	0.455	0.496	0.351	0.529
Severe	0.560	0.358	0.385	0.266	0.437
Drop vs Normal (Mild)	-2%	-4%	-4%	-5%	-3%
Drop vs Normal (Medium)	-6%	-15%	-16%	-18%	-12%
Drop vs Normal (Severe)	-17%	-33%	-35%	-38%	-27%

A clear monotonic degradation trend can be observed across all metrics as illumination conditions worsen. Even under mild degradation, performance drops are noticeable. Under severe degradation, mAP50 decreases from 0.590 to 0.385, which corresponds to a relative reduction of approximately 35%. Similarly, mAP50–95 decreases by roughly 38%, which indicates a substantial decline in both classification confidence and quality of localization. These results confirm that illumination degradation causes a significant robustness gap in the baseline detector performance.

The results reveal a clear asymmetry between precision and recall as illumination severity increases. While precision decreases from 0.673 to 0.560 (a relative drop of approximately 17%) under severe conditions, recall decreases from 0.538 to 0.358, corresponding to a relative drop of approximately 33% — nearly twice as large as the precision drop. As illustrated in Figure 3.1, the recall curve exhibits a noticeably steeper downward trend, especially during the transitions from

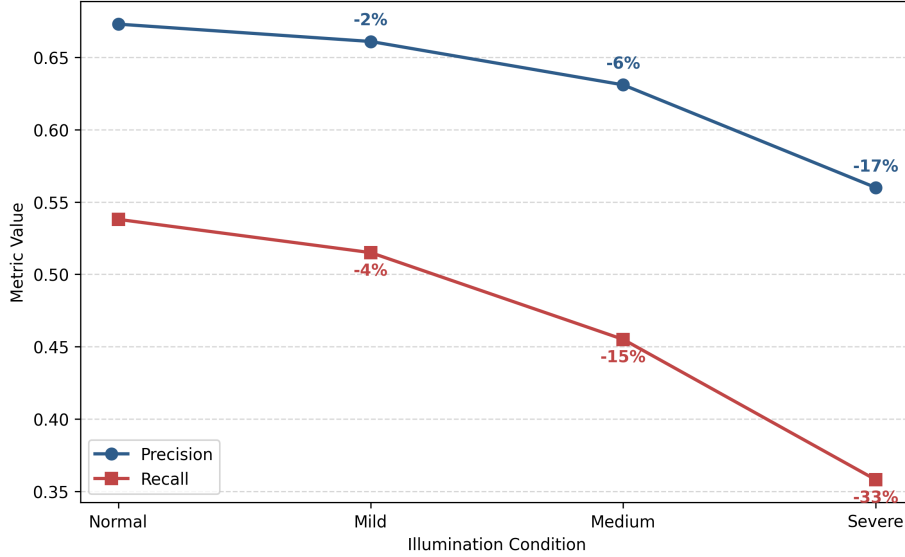


Figure 3.1: Precision and Recall under increasing illumination degradation. Recall exhibits a steeper decline, indicating growth of false negatives under low-light conditions.

Mild to Medium and from Medium to Severe degradation. The F1 score, which balances both metrics, decreases from 0.598 to 0.437 (-27%), which confirms the overall degradation trend.

This asymmetry indicates that under low-light conditions the detector increasingly fails to output detections for existing objects, so false negatives become more frequent. Such behavior corresponds to a confidence collapse effect, where object evidence becomes weaker and fewer predictions surpass the confidence threshold.

As shown in Table 3.2, the relative degradation in mAP50–95 is stronger than that in mAP50 (38% versus 35%). Since mAP50–95 measures performance under stricter IoU thresholds, this trend shows that bounding box localization becomes less accurate and less stable under reduced light. This indicates that low-light conditions affect not only the recognition of object presence, but also the precision of predicted bounding boxes.

Based on the current analysis, the dominant failure modes under low-light conditions can be summarized as:

- **Missed detections (false negatives)** caused by reduced confidence and weaker object evidence. This is the dominant failure mode, with recall degrading approximately twice as fast as precision under severe illumination degradation;
- **Localization instability**, shown in the stronger drop of mAP50–95, compared to mAP50, indicating that bounding box predictions become less accurate under reduced light.
- **Non-linear sensitivity** to illumination severity, where moderate degradation is tolerated but stronger intensity compression causes a strong performance drop.

4 Methodology

Based on the failure mode analysis in Section 3, we observed that recall degrades substantially faster than precision under increasing illumination severity. This indicates that the dominant failure mode is the growth of false negatives rather than a surge in false positives. Such behavior suggests that under low-light conditions the detector becomes increasingly conservative, potentially due to systematic suppression of confidence scores.

4.1 Curriculum-Based Robustness Training (CBRT)

We propose a training-level robustness strategy based on curriculum learning principles [3]. Curriculum learning demonstrates that presenting training examples in a meaningful order can improve convergence and generalization compared to random order or uniform exposure.

The observed degradation pattern is non-linear, with performance collapsing more rapidly under stronger illumination compression. This suggests that abrupt exposure to severe degradation may introduce excessive domain shift during training.

We propose a staged fine-tuning procedure:

- Stage 1: Fine-tuning with Normal and Mild conditions,
- Stage 2: Fine-tuning including Medium degradation,
- Stage 3: Fine-tuning including Severe degradation.

At each stage, training data from all previous degradation levels is retained to prevent forgetting of features the model learned under milder conditions. Degraded training images are pre-generated using the transformation model defined in Eq. (4).

To isolate the effect of curriculum approach, we additionally train a naive augmentation baseline in which all four degradation levels (Normal, Mild, Medium, Severe) are mixed uniformly from the start of training for the same total number of epochs. This comparison allows us to determine whether the CBRT provides a measurable benefit over unstructured augmentation with the same data.

While prior works have explored illumination-related learning strategies, our approach differs in methodology. MAET [7], for example, addresses low-light detection through a self-supervised multitask framework that jointly decodes degradation parameters and object predictions via an additional decoder and orthogonal regularity loss, with degradation severities randomly sampled from fixed distributions throughout training. Meanwhile CBRT does not require

architectural changes to the detector and explicitly orders degradation exposure from mild to severe across distinct training stages, instead of mixing all severities uniformly from the start. The naive augmentation baseline described above mirrors this uniform-exposure strategy and serves as a direct comparison point.

4.2 Illumination-Conditioned Confidence Calibration (ICCC)

Under illumination degradation, feature activations may be weakened due to dynamic range compression and reduced contrast. As a consequence, predicted confidence scores may shift downward even if the object localization remains approximately correct. This leads to an increased number of detections falling below the fixed confidence threshold, therefore increasing false negatives.

Our hypothesis is that adjusting the confidence threshold under low-light conditions can improve the results. Specifically, a moderate reduction of the confidence threshold may improve recall without causing a significant increase in false positives.

Let τ_0 denote the default confidence threshold used in the baseline detector. We introduce an illumination-conditioned threshold:

$$\tau(\gamma) = \tau_0 - k \cdot g(\gamma), \quad (5)$$

where γ represents the illumination severity level, k is a calibration coefficient and $g(\gamma)$ is a monotonic function reflecting degradation strength.

We define $g(\gamma) = \gamma - 1$, which results in zero adjustment under normal conditions ($\gamma = 1$) and increases linearly as conditions worsen. This simple linear mapping serves as a baseline, and more complex functions can be explored during future work.

The calibration coefficient k controls the sensitivity of threshold adjustment. We determine k using grid search, and we select the value that maximizes the F1-score (Eq. (3)) under each degradation condition. We use F1-score as the optimization objective because ICCC involves a direct trade-off between recall and precision: lowering the confidence threshold recovers missed detections, which improves recall but may also allow additional false positives, which would hurt precision. The F1-score balances both metrics in a single value, ensuring that the selected k improves recall without disproportionately degrading precision.

A key requirement of ICCC is the knowledge of the illumination degradation parameter γ . In our experimental setup, this value is known by design since we apply degradation synthetically. In real-world scenarios, γ must be estimated from the input image itself. We use the mean grayscale

intensity $\bar{I}$ of the input image, normalized to $[0, 1]$. The choice is motivated by the structure of the degradation model in Eq. (4): gamma correction with $\gamma > 1$ and brightness scaling with $\alpha < 1$ both systematically reduce the mean intensity of the image. The mean intensity is therefore a monotonically decreasing function of γ , which allows for a simple parametric fit. We fit a second-degree polynomial:

$$\hat{\gamma}(\bar{I}) = a \cdot \bar{I}^2 + b \cdot \bar{I} + c \quad (6)$$

The fitting procedure and estimation quality on the calibration split are reported in Section 5.2.1.

To evaluate the sensitivity of ICCC to estimation errors, we additionally assess a variant where the true degradation parameter γ is used instead of an estimated one.

The full ICCC inference pipeline for an input image I proceeds as follows:

- 1 The mean grayscale intensity $\bar{I}$ is computed and passed through the fitted polynomial to obtain $\hat{\gamma}$.
- 2 The adaptive threshold is computed as $\tau(\hat{\gamma}) = \tau_0 - k^* \cdot (\hat{\gamma} - 1)$ and is clipped to $[0.01, 1.0]$ to ensure numerical validity.
- 3 The baseline YOLOv8s model is run on the image with the adaptive threshold as its confidence filter.
- 4 The filtered detections are returned as the final output.

This pipeline does not modify the model weights, the architecture, or the preprocessing of the input image - it operates entirely as a post-processing step on the detector’s raw predictions. The only added cost is the computation of mean grayscale intensity and one polynomial evaluation per image.

Despite the fact that confidence threshold tuning is a well-established post-processing technique in object detection, the contribution of ICCC lies not only in threshold adjustment itself but in conditioning the threshold based on illumination severity, therefore making it adaptive to degradation-specific confidence suppression patterns.

4.3 Comparative and Combined Evaluation

The two proposed strategies address robustness at different levels:

- ICCC operates at the decision level by compensating for confidence bias;

- CBRT operates at the representation level by improving feature robustness.

Since these methods address robustness at different levels of the pipeline, we evaluate them both independently, after that, we investigate a combined strategy incorporating both curriculum-based training and confidence calibration to determine whether complementary improvements can be achieved.

5 Configuration for CBRT and ICCC

This section describes the configuration used to evaluate the proposed robustness improvement strategies introduced in Section 4. Three models are involved in the evaluation: the baseline detector, a model fine-tuned with the proposed Curriculum-Based Robustness Training (CBRT), and a naive augmentation baseline. The Illumination-Conditioned Confidence Calibration (ICCC) method is applied at inference time and does not require additional training. To ensure fair comparison, all three models share the same training configuration as the baseline, the same Ultralytics pretrained initialization, and the same random seed. The only difference between them is the data scheduling strategy.

5.1 CBRT Training Configuration

We implement CBRT as the three-stage fine-tuning procedure defined in Section 4.1.

In order to perform curriculum-based exposure, three additional versions of the COCO 2017 training set were generated by applying the controlled low-light transformation defined in Eq. (4) to each training image. We use the same degradation parameters described in Table 3.1. The bounding box annotations are identical across all degradation levels, as the geometric structure of each image is preserved by the transformation. Degraded images are pre-generated before training.

CBRT consists of three sequential stages:

- Stage 1 (8 epochs): Training data is composed of Normal and Mild images in equal proportion (50/50);
- Stage 2 (8 epochs): Training data includes Normal, Mild, and Medium images in equal proportion (1/3 each);
- Stage 3 (9 epochs): Training data includes all four illumination conditions: Normal, Mild, Medium, and Severe in equal proportion (25/25/25/25).

The total number of training epochs across all stages is 25, which matches the baseline training. Together with the unchanged dataset size of 118,287 images per stage, this ensures that both the baseline and CBRT models observe approximately 2.95 million image samples in total over the course of training.

To enforce the desired proportion of degradation levels at each stage, the training data was specified through explicit file lists rather than the built-in Ultralytics fraction parameter. The default fraction mechanism selects samples deterministically from the beginning of the dataset

in alphabetical order, which would lead to systematic class imbalance when combining multiple degradation folders. In order to avoid this bias, we generate three text files, each containing exactly 118,287 absolute image paths sampled uniformly at random seed 42 from the corresponding combination of degradation levels. Each stage uses its own list and a corresponding configuration file pointing to it. Each subsequent stage is initialized from the last weight produced by the previous stage.

The naive augmentation baseline is trained for 25 epochs in a single run with the same Normal+Mild+Medium+Severe (25/25/25/25) data mix used in CBRT Stage 3, the same total training budget, and the same configuration. The only difference from CBRT is the absence of curriculum ordering: all four severity levels are presented uniformly from the first epoch.

5.2 ICCC Configuration

The setup for ICCC consists of four components: a validation set partition, an automatic γ estimator that calculates illumination severity from image content, a grid search procedure for selecting the calibration coefficient k , and the final inference pipeline that combines these components into an adaptive thresholding scheme.

Since the COCO test set annotations are not publicly available, the COCO 2017 validation set serves both as calibration and final evaluation data. To prevent the calibration procedure from overfitting the held-out evaluation, the validation set is partitioned into two disjoint subsets:

- Calibration split: 20% of validation set (1000 images), used to fit the γ estimator and to select the calibration coefficient k via grid search.
- Holdout split: 80% of validation set (4000 images), reserved for final evaluation.

We perform the split by deterministic random sampling with random seed 42 to ensure reproducibility. The two subsets do not overlap, and the same partition is used across all illumination conditions: each image in the calibration split has corresponding versions under Normal, Mild, Medium, and Severe degradation, generated using the same transformation as in Eq. (4) with the parameters from Table 3.1.

5.2.1 Automatic γ estimation

We fit the polynomial from Eq. 6 using ordinary least squares via the `numpy.polyfit` routine. The fitting data consists of pairs $(\bar{I}_i, \gamma_i)$ collected from the calibration split: for each of the 1000

calibration images, the mean intensity is computed under each of the four illumination conditions, yielding 4000 data points in total.

The fitted polynomial achieves a mean absolute error (MAE) of 0.18 on the calibration data. The per-condition prediction quality is summarized in Table 5.1.

Table 5.1: Per-condition γ estimation quality on the calibration split.

Condition	True γ	Predicted $\hat{\gamma}$ (avg)	MAE
Normal	1.00	1.14	0.15
Mild	1.40	1.48	0.19
Medium	1.80	1.78	0.16
Severe	2.20	1.99	0.22

The estimator captures the overall trend correctly: the mean predicted $\hat{\gamma}$ increases monotonically with the true γ . A systematic underestimation is observed under severe degradation, where the predicted $\hat{\gamma}$ averages 1.99 instead of 2.20. This effect is a consequence of the reduced dynamic range of highly compressed images - under severe gamma correction with $\gamma = 2.2$ and brightness scaling with $\alpha = 0.40$, the usable intensity range collapses and the mean intensities of different images are pushed close together, which makes Medium and Severe conditions less separable in the feature space. We accept this limitation as a known characteristic of the estimator rather than introducing additional complexity, since the resulting threshold deviation is small (approximately 0.02 in absolute terms) and does not significantly change the behavior of ICCC, which we confirm by comparing the results of ICCC performance with true γ and estimated γ in Section 6.2.

5.2.2 Calibration of coefficient k

The calibration coefficient k in the adaptive threshold formula from Eq. (5) is selected via grid search on the calibration split.

The grid covers seven values: $k \in \{0.00, 0.02, 0.05, 0.08, 0.10, 0.12, 0.15\}$, where $k = 0$ corresponds to a non-adaptive baseline with constant threshold τ_0 .

For each value of k in the grid and each illumination condition (Normal, Mild, Medium, Severe), the adaptive threshold is computed using the true γ of the condition, since γ is known by construction at calibration time. This design decision isolates the calibration of k from the accuracy of the γ estimator: residual errors in γ estimation are not absorbed into k . The baseline YOLOv8s model is then evaluated on the calibration split under the corresponding illumination condition via `model.val(conf= $\tau$ )`, with $\tau_0 = 0.25$ as the standard inference threshold and default

IoU threshold 0.7 for non-maximum suppression. Precision, Recall, and F1-score are recorded for each run. As was stated in Section 4.2 the optimization objective is the average F1-score across the four conditions.

The grid search results are shown in Table 5.2. The maximum average F1-score of 0.5235 is achieved at $k^* = 0.10$, which is adopted as the calibrated value for the baseline model.

Table 5.2: Average F1-score across illumination conditions on the calibration split as a function of k . The best results are highlighted in bold.

k	avg F1 $\uparrow$	Normal	Mild	Medium	Severe
0.00	0.5166	0.5847	0.5636	0.5097	0.4085
0.02	0.5184	0.5847	0.5646	0.5136	0.4106
0.05	0.5212	0.5847	0.5641	0.5163	0.4197
0.08	0.5225	0.5847	0.5629	0.5175	0.4249
0.10	0.5235	0.5847	0.5624	0.5187	0.4281
0.12	0.5209	0.5847	0.5619	0.5142	0.4226
0.15	0.5220	0.5847	0.5627	0.5153	0.4252

Several observations are notable. First, the F1-score on the Normal condition remains constant across all values of k , since $\tau(\gamma = 1) = \tau_0$ regardless of k - this confirms that ICCC by design does not modify detector behavior under normal illumination. Second, the magnitude of improvement grows with degradation severity after the Mild condition: at $k^* = 0.10$ relative to $k = 0$, the F1-score gain is negative (-0.21%) on Mild, $+1.77\%$ on Medium, and $+4.8\%$ on Severe conditions. This scaling makes sense because confidence suppression becomes more pronounced under stronger illumination degradation, and therefore threshold compensation results in larger gains. Third, the average F1-score exhibits a clear optimum at $k^* = 0.10$ with a decline at higher values, indicating that excessive threshold reduction begins to admit too many false positive detections.

In the experiments reported in Section 6, the adaptive threshold is computed by averaging $\hat{\gamma}$ across all images of a given evaluation condition rather than evaluating per-image, since Ultralytics’ validation accepts a single confidence value per validation run. This approximation is justified by the fact that within each evaluation condition the true illumination severity is constant, and the estimator’s variance arises mainly from inter-image content variability rather than from inter-condition variation.

5.3 Combined CBRT and ICCC setup

Since ICCC operates at inference time and does not require additional training, the combined strategy requires no modifications to the CBRT model. The same inference pipeline de-

scribed in Section 4.2 is applied on top of the CBRT model, with the only difference being a separate calibration of the coefficient k specific to the CBRT model.

The separate calibration is necessary because the CBRT model was trained on degraded data across all illumination conditions, which may result in different confidence score distributions compared to the baseline detector. A coefficient k calibrated on the baseline model may therefore not transfer directly to CBRT.

The grid search procedure from Section 5.2.2 is repeated with the CBRT model weights substituted for the baseline weights. The same calibration split, degradation conditions, grid of k values, and F1-score optimization objective are used. The results are shown in Table 5.3.

Table 5.3: Average F1-score across illumination conditions on the calibration split as a function of k for the CBRT model.

k	avg F1	Normal	Mild	Medium	Severe
0.00	0.5410	0.5632	0.5566	0.5377	0.5066
0.02	0.5402	0.5632	0.5558	0.5377	0.5042
0.05	0.5395	0.5632	0.5555	0.5338	0.5053
0.08	0.5399	0.5632	0.5565	0.5351	0.5050
0.10	0.5393	0.5632	0.5541	0.5350	0.5049
0.12	0.5401	0.5632	0.5573	0.5350	0.5049
0.15	0.5396	0.5632	0.5555	0.5350	0.5049

The grid search yields $k_{\text{CBRT}}^* = 0.00$, indicating that confidence threshold adjustment provides no benefit when applied on top of the CBRT model. This result is discussed in detail in Section 6.3.

6 Results

This section presents the experimental results for the two proposed robustness improvement strategies: Curriculum-Based Robustness Training (CBRT) and Illumination-Conditioned Confidence Calibration (ICCC).

Note on evaluation subsets: Table 6.1 is evaluated on the full COCO validation set (5000 images), while Table 6.2 is evaluated on the held-out subset (4000 images) reserved during the ICCC calibration procedure described in Section 5.2. The baseline detector is re-evaluated on the held-out subset for a direct comparison with ICCC.

6.1 CBRT and naive augmentation performance

Table 6.1 reports detection metrics for the baseline detector, the CBRT model, and the naive augmentation baseline across all four illumination conditions.

Table 6.1: Detection performance of Baseline, CBRT, and Naive augmentation models across illumination conditions on the full COCO val2017 set (5000 images).

Condition	Model	P $\uparrow$	R $\uparrow$	mAP50 $\uparrow$	mAP50-95 $\uparrow$	F1 $\uparrow$
Normal	Baseline	0.673	0.538	0.590	0.430	0.598
	CBRT	0.651	0.511	0.562	0.404	0.572
	Naive	0.656	0.520	0.565	0.409	0.580
Mild	Baseline	0.661	0.515	0.566	0.409	0.579
	CBRT	0.650	0.507	0.554	0.396	0.569
	Naive	0.663	0.511	0.558	0.402	0.578
Medium	Baseline	0.631	0.455	0.496	0.351	0.529
	CBRT	0.636	0.486	0.529	0.374	0.550
	Naive	0.646	0.486	0.533	0.380	0.554
Severe	Baseline	0.560	0.358	0.385	0.266	0.437
	CBRT	0.613	0.443	0.479	0.330	0.514
	Naive	0.622	0.446	0.485	0.338	0.519

The results reveal a clear pattern. Under Normal and Mild conditions, both augmentation-based models perform slightly below the baseline across most metrics. This is a logical trade-off: training on a mixture of degraded and normal images distributes model capacity across multiple appearance domains, which slightly reduces performance on clean images. However, under Medium and Severe conditions, both CBRT and Naive models substantially outperform the baseline. On Severe conditions, CBRT achieves a recall of 0.443 compared to 0.358 for the baseline (+23.7% relative), and mAP50 of 0.479 compared to 0.385 (+24.4% relative).

Figure 6.1 illustrates detection performance across four metrics. The crossover between Mild and Medium conditions, visible in every panel, marks the point at which augmentation-based training begins to outperform the baseline. The divergence is most observed in recall (panel b), where the baseline falls to 0.358 under Severe degradation while CBRT and Naive maintain 0.443 and 0.446 respectively. This recall recovery translates directly into the F1-score gap shown in panel (c): CBRT achieves F1 of 0.514 compared to 0.437 for the baseline under Severe conditions (+17.6% relative), confirming that the recall gains are not offset by precision loss and that the overall detection quality under severe illumination degradation is substantially improved. Panel (d) shows that mAP50-95 also improves substantially under stronger degradation (CBRT: 0.266 up to 0.330, +24.1% relative), which provides direct evidence for the localization-robustness improvement.

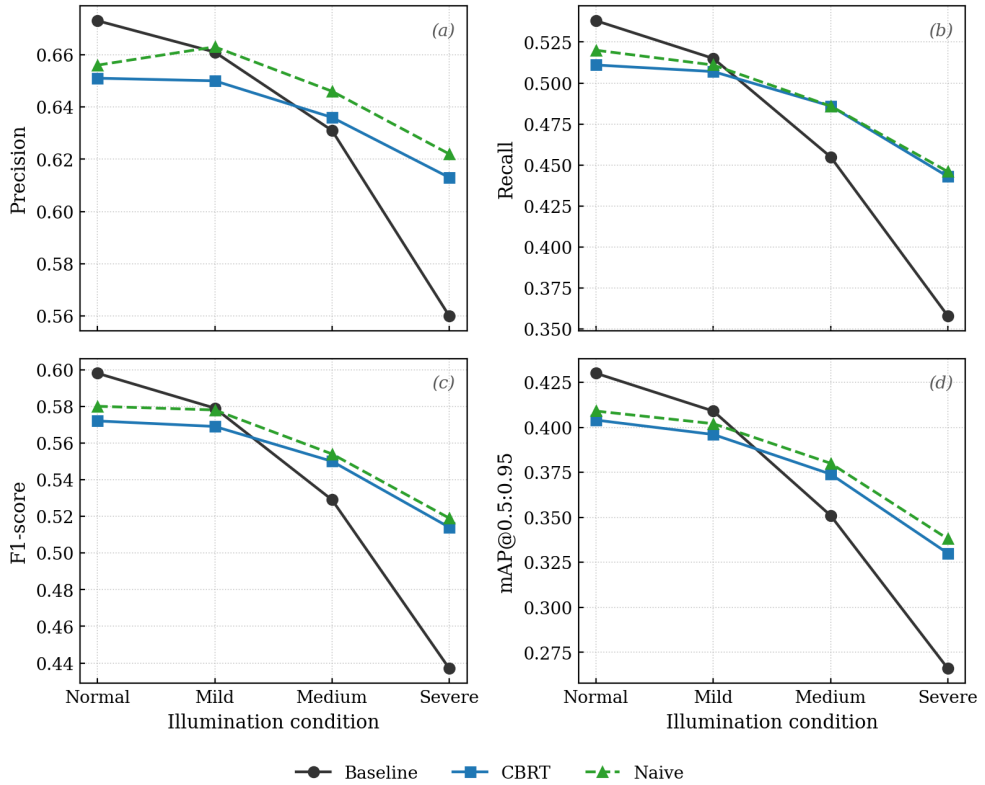


Figure 6.1: Detection performance across illumination conditions for the Baseline, CBRT, and Naive augmentation models.

Regarding the comparison between CBRT and the naive augmentation baseline, the results do not show an advantage for curriculum ordering. Under Severe conditions, the Naive model exceeds CBRT in all metrics. Under Medium conditions the Naive model similarly matches or slightly exceeds CBRT across all metrics. Notably, under Mild conditions the Naive model outperforms CBRT more visibly: F1 reaches 0.578 versus 0.569, precision reaches 0.663 versus 0.650 for CBRT, and even surpasses the baseline precision of 0.661 - the only metric on Mild conditions

where augmented model exceeds the baseline. This pattern suggests two plausible explanations. First, the staged curriculum does not introduce severe degradation examples during early training, which means the model spends the first epochs optimizing primarily for clean and mildly degraded images. When severe examples are introduced later, the model must adapt, potentially at the cost of features learned under normal conditions, so the mild forgetting of features still occurs despite the retention strategy. Second, the total training budget of 25 epochs may be insufficient for curriculum ordering to demonstrate its theoretical advantage: with more epochs, the staged exposure could allow better convergence at each difficulty level before progressing. These results show that curriculum ordering does not provide a measurable benefit over uniform exposure within our experimental setup. The main driver of robustness improvement is the inclusion of degraded data in training rather than the order in which it is introduced.

6.2 ICCC performance

All evaluations in this section use a fixed operating-point configuration ($\tau_0 = 0.25$), consistent with the ICCC calibration procedure. This configuration differs from the default Ultralytics validation settings used in Section 6.1, which sweep the full precision-recall curve to compute mAP and report P, R, and F1 at the curve’s optimal point. Since these correspond to different operating points, the absolute values of Precision, Recall, and F1 for the baseline in Table 6.2 differ from those in Table 6.1 and should not be compared directly. Within Table 6.2, both the Baseline and Baseline+ICCC rows are evaluated under identical settings, ensuring a fair comparison.

Table 6.2 reports the evaluation of ICCC applied to the baseline detector on the held-out subset. Since ICCC operates as a post-processing step that adjusts the confidence threshold without modifying the model’s raw predictions, the mAP metric is not informative for evaluating its effect: mAP integrates precision over the entire precision-recall curve and is invariant to the choice of operating threshold. We therefore evaluate ICCC using Precision, Recall, and F1-score at the adaptive threshold, which directly reflect performance at the actual operating point used in practice.

The results confirm the intended effect of ICCC: it consistently improves recall across all illumination conditions, with the magnitude of improvement scaling with degradation severity: +1.3% on Normal, +5.5% on Mild, +10.7% on Medium, and +17% on Severe conditions. Under Normal and Mild conditions the F1-score change is negligible (+0.000 in both cases), which indicates that the recall gains are fully offset by the corresponding precision loss at lighter degradation levels. On Severe conditions, the F1-score improves from 0.415 to 0.431 (+3.9% relative), which is

Table 6.2: Detection performance of the Baseline detector and Baseline with ICCC across illumination conditions on the held-out subset (4000 images). ΔR and $\Delta F1$ denote absolute improvements over the baseline. ICCC (true γ) uses the known degradation parameter directly; ICCC (est. γ) uses the automatic estimator.

Condition	Method	P $\uparrow$	R $\uparrow$	F1 $\uparrow$	ΔR	$\Delta F1$
Normal	Baseline	0.685	0.528	0.596	—	—
	ICCC (true γ)	0.685	0.528	0.596	+0.000	+0.000
	ICCC (est. γ)	0.673	0.535	0.596	+0.007	+0.000
Mild	Baseline	0.697	0.492	0.577	—	—
	ICCC (true γ)	0.659	0.515	0.578	+0.023	+0.001
	ICCC (est. γ)	0.650	0.519	0.577	+0.027	+0.000
Medium	Baseline	0.698	0.413	0.519	—	—
	ICCC (true γ)	0.615	0.458	0.525	+0.045	+0.006
	ICCC (est. γ)	0.617	0.457	0.525	+0.044	+0.006
Severe	Baseline	0.671	0.300	0.415	—	—
	ICCC (true γ)	0.528	0.363	0.430	+0.063	+0.015
	ICCC (est. γ)	0.559	0.351	0.431	+0.051	+0.016

particularly meaningful given that severe illumination represents the most practically challenging scenario.

To assess the contribution of the automatic γ estimator, Table 6.2 additionally reports results for ICCC with the true γ , where the actual degradation parameter is used directly instead of the estimated one. The gap between true and estimated variants is negligible across all conditions: F1 differences are within 0.001 on Normal, Mild, and Medium, and on Severe conditions estimated ICCC marginally outperforms true ICCC (F1 0.431 vs 0.430). This effect is a consequence of the estimator’s underestimation of γ on severely degraded images, which results in a less aggressive threshold reduction and better precision preservation. This confirms that the polynomial γ estimator introduces negligible performance loss compared to the true upper bound, and that the ICCC is robust to moderate estimation errors.

The behavior on Normal conditions deserves specific attention. The recall improvement of +1.3% and the precision decrease of -1.8% under Normal illumination are a direct consequence of the γ estimator’s systematic overestimation of γ on clean images (predicted $\hat{\gamma} = 1.14$ instead of true $\gamma = 1.00$, as reported in Table 5.1). This causes a small downward adjustment of the confidence threshold even for non-degraded images. The effect is negligible in practical terms ($\Delta F1 = +0.000$) but illustrates a known limitation of the estimator discussed in Section 5.2.2.

The precision-recall trade-off introduced by ICCC is consistent with its design: lowering the confidence threshold recovers detections that were previously suppressed by the confidence

collapse effect identified in Section 3.3, at the cost of admitting some additional false positives. The F1-score improvement on Medium and Severe conditions confirms that on balance, across stronger degradation levels, the recovered true positives outweigh the introduced false positives.

6.3 Combined CBRT and ICCC

The grid search for k applied to the CBRT model yields $k_{\text{CBRT}}^* = 0.00$ (Table 5.3), indicating that confidence threshold adjustment provides no benefit when applied on top of the CBRT model. Consequently, the combined CBRT+ICCC strategy is equivalent to CBRT alone, and no separate evaluation is required.

This result directly shows that CBRT and ICCC do not produce complementary improvements. The explanation lies in the nature of the failure mode they both address. As identified in Section 3.3, the dominant failure mode under low-light conditions is confidence collapse: object evidence becomes weaker under illumination degradation, causing predicted confidence scores to fall below the detection threshold. ICCC compensates for this effect at the decision level by lowering the threshold. CBRT addresses the same effect at the representation level: by training on degraded images, the model learns feature representations that remain discriminative under low-light conditions, which prevents confidence scores from collapsing in the first place. When the representation-level solution is in place, the decision-level correction becomes redundant.

This interpretation is supported by the confidence distributions implied by the grid search results. For the baseline model, increasing k from 0.00 to 0.10 improves average F1 by +1.34%, indicating that confidence scores are systematically suppressed under degradation. For the CBRT model, no value of $k > 0$ improves upon $k = 0$, indicating that confidence scores are no longer suppressed - the training-time exposure to degraded data has effectively resolved the suppression.

6.4 Comparison with Existing Methods

In this section, we compare the proposed CBRT and ICCC methods with existing approaches to low-light object detection identified in Section 2. The comparison is organized at three levels. First, we perform a direct empirical comparison against low-light image enhancement used as a preprocessing step, since this family of methods can be evaluated within our exact experimental setup: the same detector, the same validation images, and the same degradation model. Second, we evaluate MAET [7], a representative training-time method, directly against its own baseline within our setup, to see if the relative gains can be compared to those of CBRT and ICCC. Finally, since a fair numerical comparison with our methods is precluded for both MAET

and DAI-Net [8], we provide a qualitative and contextual comparison against these training-time approaches by their methodological properties.

6.4.1 Direct Comparison with Enhancement-Based Preprocessing

To enable a fair, direct comparison, we evaluate enhancement-based preprocessing within our own setup. We select three representative enhancement methods that span the main families of low-light enhancement: Zero-DCE [11], a zero-reference curve-estimation network, SCI [16], a self-calibrated illumination learning method, and CLAHE [26], a classical contrast-limited adaptive histogram equalization baseline. For each method, every validation image (under all four illumination conditions) is enhanced using the publicly available pretrained models, and the resulting enhanced images are passed to the same baseline YOLOv8s detector used throughout this work. This isolates the effect of preprocessing and allows a direct comparison with the representation-level CBRT strategy.

Table 6.3 reports the detection performance of the baseline detector with each enhancement method applied, alongside CBRT, across all four illumination conditions.

Table 6.3: Direct comparison of enhancement-based preprocessing against the proposed CBRT, all evaluated with the same baseline YOLOv8s detector on the full COCO val2017 set (5000 images). Enhancement methods modify only the input image; the detector is held fixed. The best result in each block is in bold.

Condition	Method	P↑	R↑	mAP50↑	mAP50–95↑	F1↑
Normal	Baseline	0.673	0.538	0.590	0.430	0.598
	CLAHE	0.677	0.525	0.582	0.423	0.591
	Zero-DCE	0.674	0.511	0.570	0.413	0.581
	SCI	0.657	0.478	0.536	0.385	0.554
	CBRT (ours)	0.651	0.511	0.562	0.404	0.572
Mild	Baseline	0.661	0.515	0.566	0.409	0.579
	CLAHE	0.654	0.496	0.549	0.394	0.564
	Zero-DCE	0.644	0.493	0.542	0.389	0.558
	SCI	0.641	0.472	0.522	0.374	0.543
	CBRT (ours)	0.650	0.507	0.554	0.396	0.569
Medium	Baseline	0.631	0.455	0.496	0.351	0.529
	CLAHE	0.606	0.435	0.470	0.331	0.507
	Zero-DCE	0.606	0.426	0.466	0.327	0.500
	SCI	0.600	0.410	0.454	0.319	0.487
	CBRT (ours)	0.636	0.486	0.529	0.374	0.550
Severe	Baseline	0.560	0.358	0.385	0.266	0.437
	CLAHE	0.532	0.334	0.355	0.242	0.410
	Zero-DCE	0.525	0.323	0.343	0.234	0.400
	SCI	0.524	0.318	0.340	0.231	0.396
	CBRT (ours)	0.613	0.443	0.479	0.330	0.514

The results reveal a clear and consistent pattern. Across all four illumination conditions, every enhancement method degrades detection performance relative to the unprocessed baseline. Under Normal illumination, the degradation is mild for CLAHE (mAP50 $0.590 \rightarrow 0.582$) but more pronounced for the learned methods, with SCI causing the largest drop (mAP50 $0.590 \rightarrow 0.536$). The negative effect grows under stronger degradation: on Severe conditions, all three methods reduce mAP50 below the baseline value of 0.385, with SCI and Zero-DCE falling to 0.340 and 0.343 respectively. So applying enhancement preprocessing to severely degraded images makes detection worse, not better.

CBRT shows the opposite pattern. It performs slightly worse than the baseline on Normal and Mild conditions, but clearly outperforms both the baseline and all enhancement methods on Medium and Severe. On Severe degradation, CBRT achieves mAP50 of 0.479 compared to 0.385 for the baseline and 0.355 for the best result of enhancement models. This confirms that, within our setup, modifying the detector’s training data is far more effective than attempting to restore the input image before detection.

These findings provide direct support for the observations reported by Wu et al. [25], who found that low-light image enhancement models consistently fail to improve, and often degrade, object detection performance.

A direct comparison of absolute metrics between ICCC and the enhancement methods is not appropriate, because ICCC is evaluated at a fixed operating point ($\tau_0 = 0.25$, as described in Section 6.2). To compare the methods fairly despite this difference, we report the improvement of each method relative to its own baseline, evaluated under identical settings. Since the mAP metric is uninformative for ICCC (discussed in Section 6.2), we use F1-score and recall as the common comparison metrics. Here we use ICCC with automatic γ estimator, which reflects the realistic deployment scenario.

Table 6.4 makes the contrast clear. All three enhancement methods produce negative gains in both recall and F1-score across every illumination condition, confirming that enhancement preprocessing actively harms detection. ICCC, in contrast, produces positive recall gains under all conditions, with the magnitude scaling with degradation severity (+0.007 on Normal up to +0.051 on Severe), and non-negative F1 gains throughout. The difference is at its peak under Severe degradation, where ICCC recovers recall by +0.051 and F1 by +0.016 while enhancement methods lose between 0.024 and 0.040 in recall and between 0.027 and 0.041 in F1-score. This demonstrates that a lightweight, decision-level adjustment is more effective than computationally heavier image-restoration preprocessing for improving detection in low light conditions.

Table 6.4: Comparison of relative gains over the respective baseline. For ICCC, gains are computed against the baseline at $\tau_0 = 0.25$ (Table 6.2 settings). ΔR and $\Delta F1$ denote absolute change over the corresponding baseline. Positive values indicate improvement.

Condition	Method	ΔR	$\Delta F1$
Normal	CLAHE	-0.013	-0.007
	Zero-DCE	-0.027	-0.017
	SCI	-0.060	-0.044
	ICCC (ours)	+0.007	+0.000
Mild	CLAHE	-0.019	-0.015
	Zero-DCE	-0.022	-0.021
	SCI	-0.043	-0.036
	ICCC (ours)	+0.027	+0.000
Medium	CLAHE	-0.020	-0.022
	Zero-DCE	-0.029	-0.029
	SCI	-0.045	-0.042
	ICCC (ours)	+0.044	+0.006
Severe	CLAHE	-0.024	-0.027
	Zero-DCE	-0.035	-0.037
	SCI	-0.040	-0.041
	ICCC (ours)	+0.051	+0.016

6.4.2 Direct Comparison with a Training-Time Method (MAET)

While enhancement-based methods can be applied to any detector without modification and therefore are straightforward to evaluate within our setup, training-time methods are harder to transfer. Nevertheless, to move beyond a purely qualitative comparison, we evaluate MAET [7] directly. We use the publicly released MAET COCO model: an 80-class YOLOv3 detector trained with the MAET framework and, as its reference point, a standard YOLOv3 detector trained on COCO from the MMDetection library [6]. Both models are evaluated on our four illumination conditions using the same validation images and annotations as in the rest of this work. It is important that the MAET model was trained on its own physics-based ISP degradation rather than on the photometric degradation model used here, therefore it never encountered our degradation during training, and this evaluation also measures how well the learned robustness transfers to a different degradation process. To keep the comparison fair, all relative changes are computed against MAET’s own YOLOv3 baseline rather than against the YOLOv8s baseline used in our work, since the two detectors differ in architecture.

Table 6.5 reveals a clear and consistent pattern. The MAET model performs below its YOLOv3 baseline across all four conditions, and the gap widens monotonically with degradation severity: in terms of mAP50, MAET trails the baseline by 30.7% under Normal illumination, 40.6% under Mild, 47.5% under Medium, and 51.5% under Severe degradation, with the same

Table 6.5: Direct comparison of the pretrained MAET-COCO model against its YOLOv3 baseline, evaluated on our four illumination conditions (full COCO val2017, 5000 images). MAET is evaluated zero-shot with respect to our degradation model. Δ denotes the relative change of MAET over the YOLOv3 baseline in terms of mAP50. The best result in each block is in bold.

Condition	Method	mAP50 $\uparrow$	mAP50-95 $\uparrow$	Δ (mAP50)
Normal	YOLOv3 baseline	0.566	0.337	—
	MAET	0.392	0.221	−30.7%
Mild	YOLOv3 baseline	0.540	0.318	—
	MAET	0.321	0.178	−40.6%
Medium	YOLOv3 baseline	0.469	0.272	—
	MAET	0.246	0.135	−47.5%
Severe	YOLOv3 baseline	0.363	0.204	—
	MAET	0.176	0.094	−51.5%

trend present in mAP50-95. The deficit is already substantial on undegraded images, where MAET reaches only 0.392 mAP50 against 0.566 for the baseline.

At first glance, these results appear contradictory, since MAET is supposed to improve baseline performance on low-light images, like CBRT does, and not make it worse. However, the results are consistent once the role of the degradation model is taken into account. CBRT is trained on exactly the degradation it is evaluated on, whereas MAET is trained on a substantially different physics-based degradation and is tested here out of distribution. The fact that the gap widens with severity indicates that the mismatch between the training and test degradation matters most precisely where robustness is most needed. This shows that the benefit of degraded-data training is specific to the degradation process used: robustness learned for one degradation model does not transfer automatically to another. The architectural gap reinforces this effect. MAET is built on YOLOv3, which is weaker than the YOLOv8s used in our main experiments, and its representations are additionally biased toward its own dark-image distribution, which is already visible in reduced performance on clean images.

We emphasize that these numbers should not be read as evidence that MAET is an inferior method. In its native setting - real low-light data such as ExDark, evaluated under its own degradation pipeline - MAET reports strong improvements over its baseline (see Section 6.4.3). The comparison above demonstrates only that a pretrained low-light detector does not transfer to an unseen degradation model, which is itself a practically relevant question when attempting to reuse such a method. For this reason, and because DAI-Net [8] cannot be evaluated within our setup without full retraining, we keep the remaining comparison with training-time methods qualitative.

6.4.3 Contextual Comparison with Training-Time Methods

The direct evaluation of MAET in Section 6.4.2 illustrates why a numerical comparison between these training-time methods and our own is not meaningful: the large performance gap stems primarily from the difference in training and test degradation and from the difference in detector architecture, rather than from the methods themselves. The same reasoning applies to DAI-Net [8], which is likewise evaluated on real low-light datasets such as ExDark [15] or on synthetic data from a physics-based ISP pipeline that differs substantially from our degradation model, and which also builds on YOLOv3 rather than YOLOv8s. We instead compare MAET and DAI-Net to our approaches qualitatively, by their methodological properties, in Table 6.6.

Table 6.6: Qualitative comparison of low-light object detection approaches. “Extra components” counts auxiliary networks, decoders, or loss terms beyond the base detector.

Method	Intervention level	Retraining	Extra components
Enhancement + detector [25]	Preprocessing	No*	Enhancement net
MAET [7]	Training	Yes	Decoder + 2 losses
DAI-Net [8]	Training	Yes	Decoder + 4 losses
CBRT (ours)	Training	Yes	None
ICCC (ours)	Post-processing	No	None

* The detector itself is not retrained, but the enhancement network is a separate trained model.

Table 6.6 highlights the methodological positioning of the proposed methods. Among training-time approaches, both MAET and DAI-Net introduce architectural modifications: MAET adds a degradation decoder D_{deg} and an orthogonal tangent regularity loss; DAI-Net introduces a reflectance decoder supervised by a pretrained RetinexNet and four additional loss terms. In contrast, CBRT achieves representation-level robustness without any architectural changes - only the data scheduling strategy differs from the baseline. ICCC occupies a distinct position as the only method in this comparison that requires no retraining of the detector at all, operating entirely as a post-processing step.

For additional context, the relative improvements achieved by CBRT over its baseline are of the same order of magnitude as those reported in the literature, despite our substantially simpler training procedure (25 epochs versus 273 epochs for MAET) and the absence of architectural modifications. Specifically, the +24.4% relative gain of CBRT in mAP50 on Severe degradation is comparable to the +18.0% gain reported for MAET and the +24.9% gain reported for DAI-Net on ExDark, although these gains are measured against different baselines and on different test distributions and are therefore not directly comparable.

Finally, the trend observed in our experiments, that exposure to degraded training data is the primary driver of robustness improvement, with the specific training schedule (curriculum

versus uniform mixing) being a secondary factor, is also visible in the literature, although it is not explicitly discussed there. In MAET, the gain from the orthogonal tangent regularity over plain multitask training is moderate (AP50 0.569 vs 0.534 on synthetic low COCO, a +6.5% relative improvement), while the gain from training on degraded data over training on well-lit data alone is substantially larger. This suggests that the main driver for improving low-light robustness is the inclusion of degraded data in training, rather than the specific scheduling or architectural mechanism used.

Overall, the comparison positions CBRT and ICCC as lightweight alternatives to the more elaborate methods in the literature. The direct comparison in Table 6.3 demonstrates that both proposed strategies substantially outperform enhancement-based preprocessing within an identical experimental setup, while Table 6.4 shows that ICCC is the only evaluated post-processing method to yield consistent positive gains. CBRT achieves relative improvements comparable to MAET and DAI-Net without architectural changes or auxiliary losses, and ICCC offers a unique post-processing option that requires no retraining at all - a property not shared by any of the compared methods.

7 Conclusion

This work investigates the robustness of YOLO-based object detection under illumination degradation through a failure-mode-driven approach. A controlled analysis of YOLOv8s under three levels of synthetic low-light degradation revealed three dominant failure modes: confidence suppression as recall degrades approximately twice as fast as precision, localization instability reflected in a stronger drop of mAP50–95 than mAP50, and non-linear sensitivity to illumination severity. Motivated by these findings, we proposed two mitigation strategies: Illumination-Conditioned Confidence Calibration (ICCC), which addresses confidence suppression at the decision level via adaptive threshold adjustment, and Curriculum-Based Robustness Training (CBRT), which addresses the same problem at the representation level through progressive exposure to degraded data during fine-tuning.

ICCC improved recall under degraded conditions with gains scaling with severity, while CBRT substantially improved both recall and localization accuracy under medium and severe degradation. However, the combination of the two methods did not yield further improvements: the calibration coefficient selected for the CBRT model was $k^* = 0$, indicating that confidence threshold adjustment becomes redundant once the representation is robust. This suggests that confidence collapse can be addressed equivalently at either level of the pipeline, and that decision-level and representation-level approaches act as alternative rather than complementary solutions to the same failure mode. A secondary finding was that curriculum ordering did not outperform uniform exposure to degraded data, indicating that the principal driver of robustness improvement is the inclusion of degraded data in training rather than the order in which it is presented. Compared with existing approaches, the proposed methods achieve relative improvements of the same order of magnitude as those reported in the literature for MAET and DAI-Net, without architectural modifications or auxiliary losses. Likewise, a direct comparison against enhancement-based preprocessing showed that restoring the input image with Zero-DCE, SCI, or CLAHE consistently degraded detection rather than improving it, confirming that modifying the detector is more effective than enhancing its input under low light. A direct evaluation of the pretrained MAET model further showed that robustness learned for one degradation model does not transfer to another: when applied to our degradation, MAET performed below its own baseline across all conditions, indicating that the benefit of degraded-data training is specific to the degradation process it was trained on.

Several limitations of this work define natural directions for future research. The evaluation is performed on synthetically degraded COCO images rather than real low-light datasets such

as ExDark, so generalizability to real-world low-light conditions remains to be verified. The γ estimator used in ICCC relies on mean grayscale intensity, which exhibits limited separability between medium and severe degradation, so more informative features or a learned estimator could improve calibration accuracy. Finally, the curriculum strategy was evaluated in a single three-stage configuration, and alternative designs may result in different conclusions about the role of ordering.

8 Use of generative AI

This chapter explains how generative artificial intelligence (AI) models were used in this work in order to follow the main guidelines of the HSE University. Claude [2], a large language model developed by Anthropic, acted as a supporting instrument for the following tasks:

1 Explanation of technical concepts. Several concepts underlying this work were initially unfamiliar and had to be studied in order to understand their purpose and correct application. Although the original papers and documentation were useful, some ideas remained unclear and were additionally explained by the model through illustrative examples and step-by-step breakdowns of the relevant formulas. This applied in particular to:

- the distinction between evaluating at a fixed operating point and sweeping the full precision-recall curve, and why this affects how mAP and F1 should be compared across methods;
- the photometric degradation model used to simulate low-light conditions, including the role of gamma correction, brightness scaling, and additive noise.
- the object detection evaluation metrics, in particular the difference between mAP50 and mAP50–95, the meaning of the IoU threshold, and the role of non-maximum suppression and the confidence threshold at the operating point.

In each case, the explanations were used only to build understanding, after which the corresponding theory was verified against the original sources cited in this work.

2 Code construction and debugging. The model helped to clarify the correct usage of several Python functions and to identify the main errors that caused scripts to crash. It was especially useful during the initial setup of the experimental pipeline: the COCO dataset had to be converted into the annotation format expected by the Ultralytics YOLO framework before YOLOv8s could be trained and evaluated, and the model helped to diagnose the formatting and configuration issues that initially prevented training from starting. It was also used to debug the grid-search and evaluation scripts and to interpret error messages produced by the framework.

3 Spelling, grammar, and punctuation. The final text of the coursework was checked for grammatical, spelling, and punctuation mistakes.

All suggestions provided by the model were verified against articles and academic papers, and underwent manual correction and refinement. They were not copied blindly, but were studied to ensure understanding of the materials and methods used.

References

- [1] Fadwa Alrowais, Saud S Alotaibi, Fahd N Al-Wesabi, Noha Negm, Rana Alabdan, Radwa Marzouk, Amal S Mehanna, and Mesfer Al Duhayyim. “Deep transfer learning enabled intelligent object detection for crowd density analysis on video surveillance systems”. In: *Applied Sciences* 12.13 (2022), p. 6665.
- [2] Anthropic. *Claude*. Large language model. 2026. URL: <https://claude.ai> (visited on May 31, 2026).
- [3] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. “Curriculum learning”. In: *Proceedings of the 26th annual international conference on machine learning*. 2009, pp. 41–48.
- [4] Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao. “Yolov4: Optimal speed and accuracy of object detection”. In: *arXiv preprint arXiv:2004.10934* (2020).
- [5] Yingfeng Cai, Tianyu Luan, Hongbo Gao, Hai Wang, Long Chen, Yicheng Li, Miguel Angel Sotelo, and Zhixiong Li. “YOLOv4-5D: An effective and efficient object detector for autonomous driving”. In: *IEEE Transactions on Instrumentation and Measurement* 70 (2021), pp. 1–13.
- [6] Kai Chen, Jiaqi Wang, Jiangmiao Pang, Yuhang Cao, Yu Xiong, Xiaoxiao Li, Shuyang Sun, Wansen Feng, Ziwei Liu, Jiarui Xu, et al. “MMDetection: Open mmlab detection toolbox and benchmark”. In: *arXiv preprint arXiv:1906.07155* (2019).
- [7] Ziteng Cui, Guo-Jun Qi, Lin Gu, Shaodi You, Zenghui Zhang, and Tatsuya Harada. “Multitask aet with orthogonal tangent regularity for dark object detection”. In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2021, pp. 2553–2562.
- [8] Zhipeng Du, Miaojing Shi, and Jiankang Deng. “Boosting object detection with zero-shot day-night domain adaptation”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 12666–12676.
- [9] Mark Everingham, Luc Van Gool, Christopher KI Williams, John Winn, and Andrew Zisserman. “The pascal visual object classes (voc) challenge”. In: *International journal of computer vision* 88.2 (2010), pp. 303–338.
- [10] Rafael C. Gonzalez and Richard E. Woods. *Digital Image Processing*. 3rd ed. Upper Saddle River, NJ: Pearson, 2008.

- [11] Chunle Guo, Chongyi Li, Jichang Guo, Chen Change Loy, Junhui Hou, Sam Kwong, and Runmin Cong. “Zero-reference deep curve estimation for low-light image enhancement”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020, pp. 1780–1789.
- [12] Wang Hao and Song Zhili. “Improved mosaic: Algorithms for more complex images”. In: *Journal of Physics: Conference Series*. Vol. 1684. 1. IOP Publishing. 2020, p. 012094.
- [13] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. *Ultralytics YOLOv8*. Version 8.0.0. 2023. URL: <https://github.com/ultralytics/ultralytics> (visited on Feb. 26, 2026).
- [14] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. “Microsoft coco: Common objects in context”. In: *European conference on computer vision*. Springer. 2014, pp. 740–755.
- [15] Yuen Peng Loh and Chee Seng Chan. “Getting to know low-light images with the exclusively dark dataset”. In: *Computer vision and image understanding* 178 (2019), pp. 30–42.
- [16] Long Ma, Tengyu Ma, Risheng Liu, Xin Fan, and Zhongxuan Luo. “Toward fast, flexible, and robust low-light image enhancement”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022, pp. 5637–5646.
- [17] Claudio Michaelis, Benjamin Mitzkus, Robert Geirhos, Evgenia Rusak, Oliver Bringmann, Alexander S Ecker, Matthias Bethge, and Wieland Brendel. “Benchmarking robustness in object detection: Autonomous driving when winter is coming”. In: *arXiv preprint arXiv:1907.07484* (2019).
- [18] Lukáš Neumann, Michelle Karg, Shanshan Zhang, Christian Scharfenberger, Eric Piegert, Sarah Mistr, Olga Prokofyeva, Robert Thiel, Andrea Vedaldi, Andrew Zisserman, et al. “Nightowls: A pedestrians at night dataset”. In: *Asian Conference on Computer Vision*. Springer. 2018, pp. 691–705.
- [19] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. “You only look once: Unified, real-time object detection”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016, pp. 779–788.
- [20] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. “Faster r-cnn: Towards real-time object detection with region proposal networks”. In: *Advances in neural information processing systems* 28 (2015).

- [21] Dimitrios Sakkos, Hubert PH Shum, and Edmond SL Ho. “Illumination-based data augmentation for robust background subtraction”. In: *2019 13th International Conference on Software, Knowledge, Information Management and Applications (SKIMA)*. IEEE. 2019, pp. 1–8.
- [22] GSDMA Sreenu and Saleem Durai. “Intelligent video surveillance: a review through deep learning techniques for crowd analysis”. In: *Journal of Big Data* 6.1 (2019), pp. 1–27.
- [23] Dai Quoc Tran, Armstrong Aboah, Yuntae Jeon, Maged Shoman, Minsoo Park, and Seunghye Park. “Low-light image enhancement framework for improved object detection in fisheye lens datasets”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 7056–7065.
- [24] Chien-Yao Wang, Hong-Yuan Mark Liao, Yueh-Hua Wu, Ping-Yang Chen, Jun-Wei Hsieh, and I-Hau Yeh. “CSPNet: A new backbone that can enhance learning capability of CNN”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*. 2020, pp. 390–391.
- [25] Xu Wu, Zhihui Lai, Zhou Jie, Can Gao, Xianxu Hou, Ya-nan Zhang, and Linlin Shen. “Low-Light Enhancement Effect on Classification and Detection: An Empirical Study”. In: *arXiv preprint arXiv:2409.14461* (2024).
- [26] Karel Zuiderveld. “Contrast limited adaptive histogram equalization”. In: *Graphics gems IV*. 1994, pp. 474–485.